

Luis Felipe García Arias

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PROFILE

I am an engineer with a research background in electronics, signal processing, and software development. I have developed prototypes from early concept through circuit and mechanical design to field testing. I adapt quickly to each project's needs, moving from PCB design, mechanical design, and firmware to DevOps, VR/game development in C#, and signal processing and machine learning as needed. Currently, I am seeking a position driven by innovation and focused on improving people's lives through technology, a goal that has guided my research in rehabilitation and assistive devices.

SKILLS & CERTIFICATIONS

Professional Skills: Rapid prototyping, Hardware–software integration, Field & user testing with patients, Clinical study protocols & risk assessment (WMO/non-WMO, METc), Technical project leadership, Code review & Git workflows, Cross-disciplinary collaboration (engineering–clinical), Scientific writing & patenting.

Programming & Tools: C++/C#, Unit Testing, Git, Linux, Docker, Python, MATLAB, Altium Designer, SolidWorks.

Certifications: Certified SOLIDWORKS Professional – Core (2016)

Languages: English C1 (TOEFL iBT), Spanish (native), Portuguese B2 (CELPE-Bras), Dutch (A2).

EXPERIENCE

2022/Jul – Present: Doctoral Researcher

University of Groningen, The Netherlands

- Developed a VR game for neck-strengthening exercises; tested with 40 healthy participants, followed by a feasibility study with 4 outpatients living with cervical dystonia.
- Co-wrote the study protocols and ethics applications for the participant studies (WMO, assessed by the METc UMCG, and non-WMO), including risk assessment and informed-consent materials.
- Implemented and unit-tested C# game logic for the rehabilitation applications, using Git for version control.
- Co-developed a video game for knee rehabilitation using EMG and IMU sensing to steer gameplay.
- Trained physics-informed neural networks on a self-built dataset for hand gesture recognition: pipeline from data-preprocessing of EMG signals up to hyperparameter optimization.

2020/Mar – 2022/Jun: Technical Project Lead

University of Groningen, The Netherlands

- Led a 5-person team in developing new versions of 3 web apps to replace monolithic versions previously developed in PHP.
- Updated the git workflow, code review and pull request processes of the team.
- Coordinated the development of front-end (Vue JS) and back-end (Laravel) based web apps to replace monolithic versions.
- Coordinated the deployment of Docker containers of the modules and database, as well as their continuous deployment.

2020/Oct – 2021/Apr: Research and Development Systems Engineer

Universidad Nacional de Colombia, Colombia

- Conducted the basic engineering for a Support system for early warnings of potential infections using data analysis techniques.
- Developed prototypes of customized assistive devices for human-machine interaction.

2017/Apr – 2020/Dec: Research and Development Systems Engineer

Universidad de Caldas, Colombia

- Co-developed an electronic system to assist physicians in measuring functional performance tests in elderly. I participated in the process of basic engineering, electronics design and assembly of the first prototype, design enclosures design for 3D printing, and coding of an integration system in python and coding of the firmware in C++.
- Built an electronic system (E-SPPB) assisting physicians in measuring functional performance in elderly patients: electronics design, 3D-printed enclosure design, and firmware/integration coding in C++ and Python, using Git for version control.
- Tested the system with 400 elderly participants across four subregions of Caldas, Colombia.
- Co-authored an application that led to a utility model patent granted by Colombia's Superintendence of Industry and Commerce (SIC), Colombia. Patent No.: NC2020/0004133.

EDUCATION

2022 – Present: PhD Researcher

University of Groningen, The Netherlands

Movement-based interactive systems for therapy: rehabilitation and assistive gaming driven by EMG/IMU sensing and physics-informed neural networks.

2020 – 2022: MSc in Engineering – Industrial Automation

Universidad Nacional de Colombia, Manizales, Colombia

2012 – 2017: Electronics Engineering & Mechatronics Engineering

Universidad Nacional de Colombia / Universidad de Caldas, Manizales, Colombia

Dual degree.

SELECTED PUBLICATIONS

Virtual Reality Supported Neck Stretching Exercises to Improve Cervical Range of Motion. *L. F. Garcia Arias, Y. Zhang, M. Tijssen, E. Wilhelm.* 2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2025. DOI: 10.1109/EMBC58623.2025.11253550.

IMU-Mouse: diseño e implementación de un dispositivo apuntador dirigido al desarrollo de interfaces adaptativas para personas con discapacidad física. 2018. DOI: 10.22430/22565337.727.

PATENTS

Utility Model Patent – electronic device for functional assessment of elderly patients (E-SPPB). Granted by Colombia's Superintendence of Industry and Commerce, 2025.